

EC-231 Operating Systems

BS(CE)-2021

LAB REPORT # 3



Submitted By:

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	Presentation (Format (0.5) + Title (0.5) + Conclusion(1) + Procedure(2) + Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

**Lab Instructor.
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Experiment #3

Basic Shell scripting using gedit/vi

Objective:

- 1.Vi Introduction
- 2.Setting access permission of files
- 3.Executing shell script
- 4.User and System Variables
- 5.Shell Arithmetic
- 6.Exit status

Software Used:

1. Ubuntu
2. gedit/vi(Text Editor)

Procedure:

Introduction to Vi:

Under Linux, there is a free version of Vi called Vim (Vi Improved). Vi (pronounced vee-eye) is an editor that is fully in text mode, which means that all actions are carried out with the help of text commands. This editor, although it may appear of little practical use at first, is very powerful and can be very helpful in case the graphical interface malfunctions.

Syntax:

- 1.vi name_of_the_file
- 2.Once the file is open, you can move around by using cursors.
- 3.Press I to switch to insert mode.
- 4.Press escape key to switch to command mode.

How to write shell script:

Following steps are required to write shell script:

- 1.Use any editor like vi or gedit to write shell script.
- 2.To close vi window press : and then given command (i.e. q ,!q or wq)
- 3.To save and close gedit window press ALT+F4
- 4.After writing shell script set execute permission for your script as follows.

Variables in Shell:

To process our data/information, data must be kept in computers RAM memory. RAM memory is divided into small locations, and each location had unique number called memory location/address, which is used to hold our data. Programmer can give a unique name to this memory location/address called memory variable or variable (Its a named storage location that may take different values, but only one at a time).

In Linux (Shell), there are two types of variable:

(1) System variables:

Created and maintained by Linux itself. This type of variable defined in CAPITAL LETTERS.

(2) User defined variables (UDV):

Created and maintained by user. This type of variable defined in lower letters.

You can see system variables by giving command like \$ set, some of the important System variables are: HOME, USER

Tasks

Task 1:

Stimulate following commands in Linux and note down the exit status:

a. \$ expr 1+3

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo `expr 1+3`
```

Output:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
4
```

b. \$ echo \$?

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo `expr $?`
```

Output:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
0
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$
```

c. \$ echo Welcome

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo Welcome
```

Output:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
Welcome
```

d. \$ echo \$?

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $?
```

Output:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
0
```

e. \$ wildwestcanwork?

CODE:

```

1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $wildwestcanwork

```

OUTPUT:

```

huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
wildwestcanwork

```

f. \$ echo\$?

CODE:

```

1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $?

```

OUTPUT:

```

huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
$

```

g. \$date

CODE:

```

1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $date

```

OUTPUT:

```

huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
date

```

h.\$ echo \$

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $?
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
$
```

i.\$ echo\$?

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $?
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
$
```

j.\$ echo\$?

CODE:

```
1 #!/bin/bash
2 n=10
3 c=test
4 echo $n
5 echo $c
6 echo $HOME
7 echo $?
```

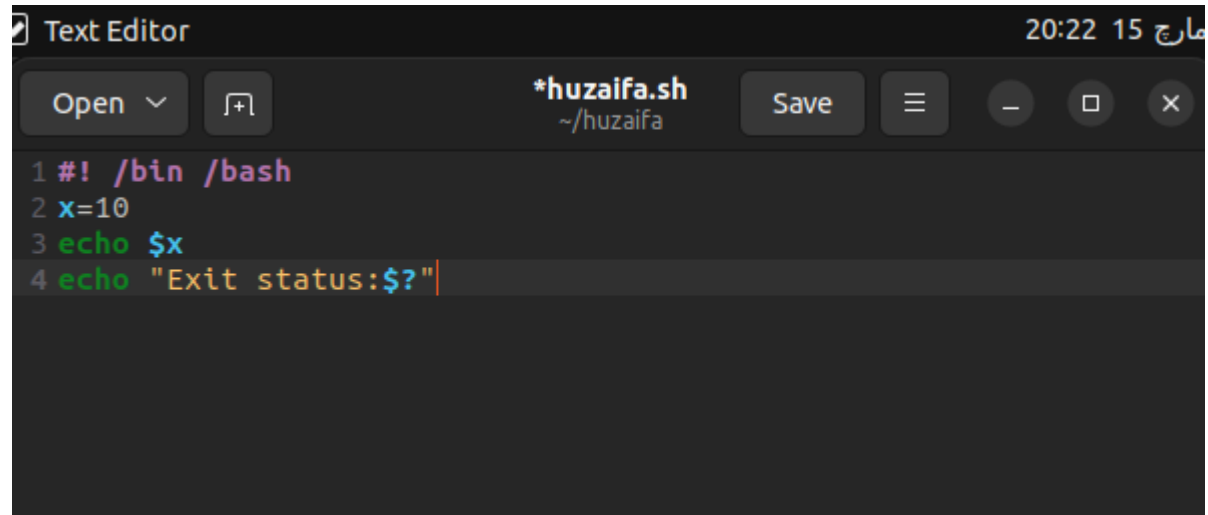
OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
test
/home/huzaifashahbaz
$
```

Task 2:

Simulate command/commands in Linux to define variable x with value 10 and print it on screen. Check exit status.

CODE:



```
1 #!/bin/bash
2 x=10
3 echo $x
4 echo "Exit status:$?"
```

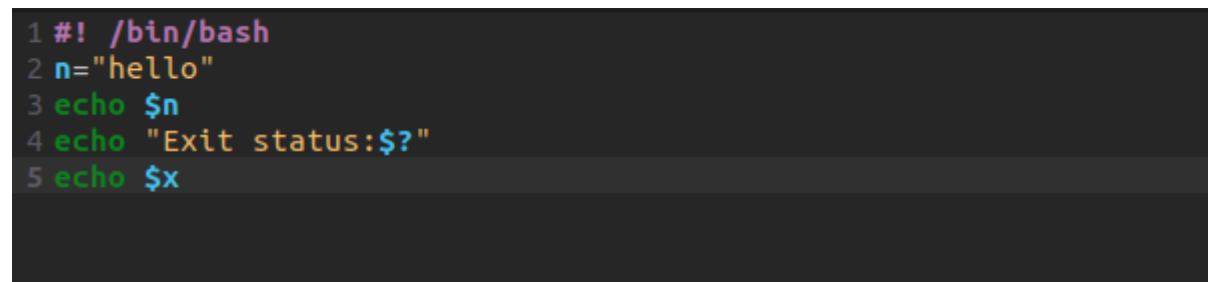
Output:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
Exit status:0
```

Task 3:

Simulate command/commands in Linux to define variable xn with value Hello and print it on screen. Check exit status.

CODE:



```
1 #!/bin/bash
2 n="hello"
3 echo $n
4 echo "Exit status:$?"
5 echo $x
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
hello
Exit status:0
```

Task 4:

Simulate command/commands in Linux to print sum of two numbers, let's say 6 and 3?

CODE:

```
1 #! /bin/bash
2 n=6
3 x=3
4 echo $n
5 echo $x
6 echo `expr 6 + 3`
```

OUTPUT:

```
huzalfashahbaz@huzalfashahbaz-VirtualBox:~/huzalfa786$ xdg-open huzalfa.sh
huzalfashahbaz@huzalfashahbaz-VirtualBox:~/huzalfa786$ ./huzalfa.sh
6
3
9
```

Task 5:

Simulate command/commands in Linux to define two variable x=20, y=5 and then to print division of x and y (i.e. x/y).

CODE:

```
1 #! /bin/bash
2 x=20
3 y=5
4 echo $x
5 echo $y
6 echo `expr 20 / 5`
```

OUTPUT:

```
huzalfashahbaz@huzalfashahbaz-VirtualBox:~/huzalfa786$ xdg-open huzalfa.sh
huzalfashahbaz@huzalfashahbaz-VirtualBox:~/huzalfa786$ ./huzalfa.sh
20
5
4
```

Task 6:

Simulate command/commands in Linux to modify above and store division of x and y to variable called z.

CODE:

```
1 #! /bin/bash
2 x=10
3 y=5
4 z=$(echo "scale=2; $x / $y" |bc)
5 echo $x
6 echo $y
7 echo $z
8 echo "The division of $x by $y is $z"
```


OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
10
5
2.00
The division of 10 by 5 is 2.00
```

Task 7:

Simulate command/commands in gedit to make a simple calculator in which addition, multiplication, subtraction and division on two user defined variables is calculated.

CODE:

```
1 #!/bin/bash
2 echo "Enter the first number:"
3 read num1
4 echo "Enter the second number:"
5 read num2
6 echo "Select operation:"
7 echo "1.Addition"
8 echo "2.Subtraction"
9 echo "3.Multiplication"
10 echo "4.Division"
11 read choice
12 case $choice in
13     1)result=$(echo "$num1 + $num2" |bc)
14     ;;
15     2)result=$(echo "$num1 - $num2"|bc)
16     ;;
17     3)result=$(echo "$num1 * $num2"|bc)
18     ;;
19     4)result=$(echo "scale=2; $num1 / $num2"|bc)
20     ;;
21     *)echo "Invalid choice"
22     exit 1
23     ;;
24 esac
25 echo "Results: $result"
```

OUTPUT:

```
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai786$ xdg-open huzai786.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai786$ ./huzai786.sh
Enter the first number:
6
Enter the second number:
8
Select operation:
1.Addition
2.Subtraction
3.Multiplication
4.Division
3
Results: 48
```

Task 8:

Simulate command/commands in gedit that contains today's date in script, execute the program and display its output.

CODE:

```
1 #!/bin/bash
2 today=$(date +%Y-%m-%d)
3 output=$(ls "$today")
4 echo "Output of ls command with today's date ($today):"
5 echo "$output"
```

OUTPUT:

```
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai786$ xdg-open huzai786.sh
huzaihashahbaz@huzaihashahbaz-VirtualBox:~/huzai786$ ./huzai786.sh
ls: cannot access '2024-03-16': No such file or directory
Output of ls command with today's date (2024-03-16):
```

Task 9:

Simulate command/commands in gedit that contains contents of your home directory in script, execute the program and display its output.

CODE:

```
1 #!/bin/bash
2 echo ls -
3 echo cd -
4 echo |chmod +x list_home_content.sh
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
ls -
cd -
chmod +x list_home_content.sh
```

Task 10:

Simulate command/commands in gedit that contains your present work directory in script, execute the program and display its output.

CODE:

```
1 #! /bin/bash
2 echo "Present Working Directory: $PWD"
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
Present Working Directory: /home/huzaifashahbaz/huzaifa786
```

Task 11:

Simulate the following commands in Linux and state which of the command is correct and why for printing your home directory location,

a)\$ echo \$HOME

OR

(b)\$ echo HOME

CODE:

A:

```
1 #! /bin/bash
2 $ echo $HOME
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
./huzaifa.sh: line 2: $: command not found
```

B:

```
1 #! /bin/bash
2 echo $HOME
```

OUTPUT:

```
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ xdg-open huzaifa.sh
huzaifashahbaz@huzaifashahbaz-VirtualBox:~/huzaifa786$ ./huzaifa.sh
/home/huzaifashahbaz
```

Result:

So, the part `b $ echo HOME` is the correct command for the Linux.

Conclusion:

In conclusion, we had learned how the commands of Linux perform in textfile Basic Shell scripting using gedit/vi. Also we make the calculator by performing basic operation by Linux commands.